

AUTONOMOUS AGENT TRUST ARCHITECTURE (AATA)

A Reference Framework for Securing Fleets of Autonomous Cognitive and Embodied Agents in Connected, Degraded, and Isolated Environments

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Protime Consulting Inc. — Framework Working Draft

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Document Control

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1. Executive Summary

The technology landscape is converging on a single pattern: **autonomous agents operating on trusted physical infrastructure without a human in the loop**. Cognitive agents (frontier LLMs), embodied agents (humanoid robots, autonomous vehicles), and infrastructure agents (orchestrators, SOC copilots) are being deployed into environments where continuous human supervision is physically or economically impossible — factory night shifts, contested networks, lunar and Martian operations, and air-gapped sovereign enclaves.

Every major vertically integrated AI program — power generation, silicon fabrication, orbital communications, mass logistics, and robotics — assumes this autonomy. None of them ships the layer that makes it safe: **a trust architecture for agent fleets**. Identity, attestation, capability scoping, behavioral hygiene, covert-channel monitoring, and intrinsic (architecture-level) alignment remain unsolved as an integrated system.

The Autonomous Agent Trust Architecture (AATA) is that layer. It defines **five planes** — Governance, Trust, Control, Observability, and Execution — a **seven-stage agent trust chain** from hardware root to retirement, and a **three-tier DDIL operating model** (Connected / Degraded / Isolated) that degrades gracefully from cloud-supervised operation to fully intrinsic self-governance. It is deployable today on enterprise agentic platforms and scales conceptually to the extreme case: agent colonies at interplanetary light-lag.

Core thesis: *when the supervisor is removed, security and alignment must be properties of the agent architecture itself — not services called over a network. AATA operationalizes that thesis with concrete, standards-mapped controls.*

2. Problem Statement and Context

2.1 The supervision gap

Current agent security models are borrowed from workload security: a central IdP issues credentials, a cloud policy engine evaluates requests, and human operators review alerts. All three assumptions fail for autonomous fleets. Agents self-replicate tasks across sub-agents; policy engines are unreachable at high latency or in air-gapped enclaves; and alert volume from thousands of agents exceeds any human review capacity. The result is a supervision gap that widens with every increment of autonomy.

2.2 The embodiment multiplier

Embodied agents convert software compromise into kinetic consequence. A hijacked reasoning loop in a chatbot leaks data; the same hijack in a 70 kg humanoid on a production floor is a physical safety event. Fleet-scale embodiment (thousands of units sharing a model baseline) also creates **monoculture risk**: one poisoned weight update or one viable prompt-injection chain compromises the entire fleet simultaneously.

2.3 The DDIL forcing function

Denied, Degraded, Intermittent, and Limited (DDIL) environments — contested RF, subterranean sites, maritime, lunar (~2.5 s RTT), and Martian (4–24 min light-lag) operations — make reach-back architectures physically impossible. DDIL is not an edge case; it is the design case. An architecture that remains secure when isolated is trivially secure when connected. The inverse is false.

2.4 Novel attack surface

Agent fleets introduce attack classes that traditional controls do not observe: inter-agent covert channels (Unicode steganography, timing channels in tool-call cadence), goal hijacking via poisoned context, capability escalation through sub-agent delegation chains, memory/RAG poisoning that persists across sessions, and model-supply-chain compromise of weights, adapters, and skill packages. Section 7 maps these to MITRE ATLAS and STRIDE.

3. Design Principles

#	Principle	Implication
P1	Intrinsic before extrinsic	Alignment constraints live in the agent's objective architecture (bounded objectives, constitutional policy), not solely in network-side guardrails. External controls are defense-in-depth, not the foundation.
P2	Identity is per-agent, hardware-rooted	Every agent instance holds a unique, attestable identity (SPIFFE/SVID pattern) chained to a hardware root of trust and a signed model/runtime digest. No shared service accounts.
P3	Capability, not permission	Agents receive scoped, short-lived, revocable capability tokens enumerating tools, data domains, spend ceilings, and blast radius. Default deny; escalation requires attested justification.
P4	Assume isolation	Every control must have a defined DDIL behavior: what it does when the backhaul disappears. Policy caches carry TTLs; trust duties fail over to a peer-attestation mesh.
P5	Evidence is non-negotiable	Every consequential decision is written to an immutable flight recorder before actuation. In isolation, evidence accrues to a reconciliation ledger audited on reconnect.
P6	Hygiene is autonomous	Drift detection, quarantine, and re-imaging are fleet self-functions. A compromised agent must be isolatable by its peers without human latency.
P7	Monoculture is a vulnerability	Fleet baselines are diversified (model tier, prompt hardening variants, staged rollout rings) so a single exploit chain cannot achieve simultaneous fleet-wide compromise.

4. Reference Architecture — The Five Planes

AATA organizes all functions into five planes. Planes are logical: in an enterprise footprint they map to distinct services; in an isolated footprint several planes collapse into on-agent components. Figure 1 shows the planes and primary data flows.

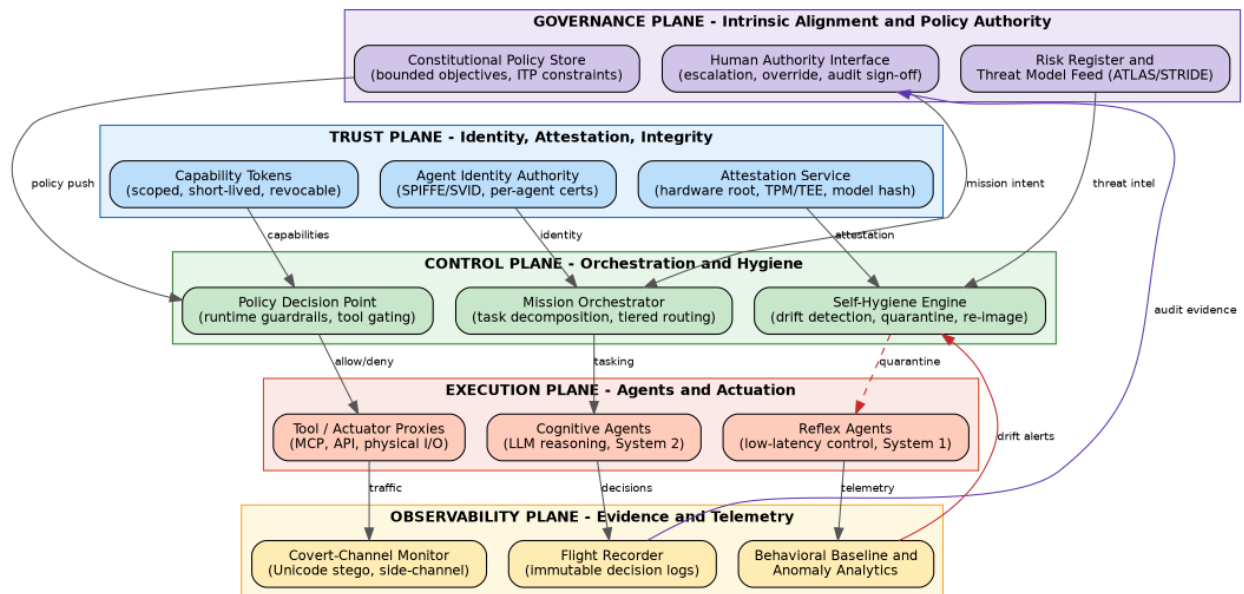


Figure 1 — AATA five-plane reference architecture with primary control and evidence flows.

4.1 Governance Plane

- **Constitutional Policy Store.** Machine-readable bounded objectives and prohibitions compiled into agent runtime constraints (ITP pattern: values as architecture, not filter). Versioned, signed, and distributed with TTLs.
- **Human Authority Interface.** The single surface for mission intent, override, escalation approval, and audit sign-off. Human-on-the-loop in Connected mode; asynchronous reconciliation authority in Isolated mode.
- **Risk Register / Threat Model Feed.** Living threat model (ATLAS/STRIDE entries, e.g., covert-channel TTPs) pushed as detection logic to the Self-Hygiene Engine and Covert-Channel Monitor.

4.2 Trust Plane

- **Agent Identity Authority.** Issues per-instance workload identities (SPIFFE SVID or equivalent X.509) bound to the attested platform. Identity lifetime is a lease, not a grant.
- **Attestation Service.** Verifies secure boot, TPM/TEE quotes, runtime SBOM, and cryptographic digests of model weights, adapters, system prompts, and skill packages before identity issuance and on re-attestation events.
- **Capability Token Service.** Mints scoped tokens: allowed tools, data classifications, network zones, actuation classes, spend/compute ceilings, and maximum delegation depth for sub-agent spawning.

4.3 Control Plane

- **Mission Orchestrator.** Decomposes intent into tasks, routes across model tiers (cost/latency/capability-aware routing), enforces delegation-depth limits, and tracks task provenance end-to-end.
- **Policy Decision Point (PDP).** Runtime gate on every tool/actuator call: evaluates capability token, constitutional constraints, and current threat posture. Fail-closed for kinetic actions, fail-degraded for informational ones.
- **Self-Hygiene Engine.** Consumes drift alerts and IOCs; executes graduated response — capability narrowing, comms isolation, quarantine, credential revocation, state wipe, and re-image from attested golden baseline.

4.4 Observability Plane

- **Flight Recorder.** Append-only, hash-chained log of prompts, retrieved context, tool calls, decisions, and actuation commands. Written pre-actuation; the evidentiary system of record.
- **Covert-Channel Monitor.** Inspects inter-agent traffic for steganographic encoding (Unicode confusables/zero-width channels), timing/side channels, and out-of-policy semantic content.
- **Behavioral Baseline and Anomaly Analytics.** Per-agent and fleet-level baselines over tool-use distributions, token economics, delegation patterns, and actuation profiles; feeds drift alerts to hygiene.

4.5 Execution Plane

- **Cognitive Agents (System 2).** LLM-based reasoning/planning components; slow, deliberate, policy-aware.
- **Reflex Agents (System 1).** Low-latency control loops (motion, safety interlocks) that cannot tolerate LLM latency and must remain safe if the cognitive layer is quarantined.
- **Tool / Actuator Proxies.** Mediated interfaces (MCP servers, APIs, physical I/O gateways). Agents never touch tools directly; every call transits a proxy that enforces the PDP verdict and emits evidence.

5. Agent Trust Chain Lifecycle

Every agent instance traverses a seven-stage lifecycle. No stage may be skipped; each produces cryptographic evidence consumed by the next. Figure 2 shows the chain.

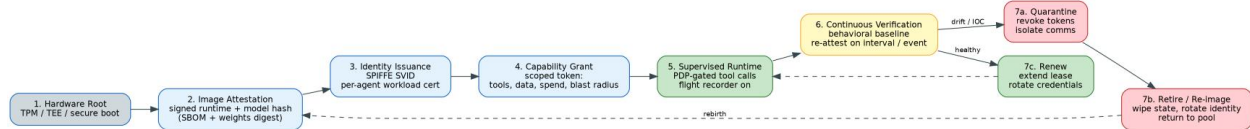


Figure 2 — Agent trust chain from hardware root through quarantine/renewal.

Stage	Name	Specification
1	Hardware Root	Secure boot with measured launch; TPM 2.0 or TEE (SEV-SNP/TDX/OP-TEE class) quote establishes platform integrity. GPU/accelerator firmware included in measurement where supported.
2	Image Attestation	Runtime SBOM plus cryptographic digests of model weights, LoRA adapters, system prompts, and skill/plugin packages verified against a signed manifest. Any mismatch blocks identity issuance.
3	Identity Issuance	SPIFFE SVID (or X.509 equivalent) minted per instance, bound to the attestation evidence. Lease-based validity (minutes–hours, footprint-dependent).
4	Capability Grant	Token enumerates: tool allowlist, data classification ceiling, network zones, actuation classes, compute/spend budget, and max delegation depth. Sub-agents inherit a strict subset (monotone attenuation).
5	Supervised Runtime	All tool/actuator calls transit proxies gated by the PDP; flight recorder captures decision context pre-actuation.
6	Continuous Verification	Re-attestation on interval, on anomaly, and on privilege escalation request. Behavioral baseline deviation scores feed the hygiene engine.
7	Disposition	Renew (healthy: rotate credentials, extend lease) Quarantine (drift/IOC: revoke tokens, isolate comms, preserve forensic state) Retire/Re-image (wipe mutable state, rotate identity, rebuild from golden baseline — the rebirth path).

6. DDIL Operating Model

AATA defines three connectivity tiers with explicit control-degradation behavior. The architecture is designed from the Isolated tier upward: every control specifies its isolated behavior first.

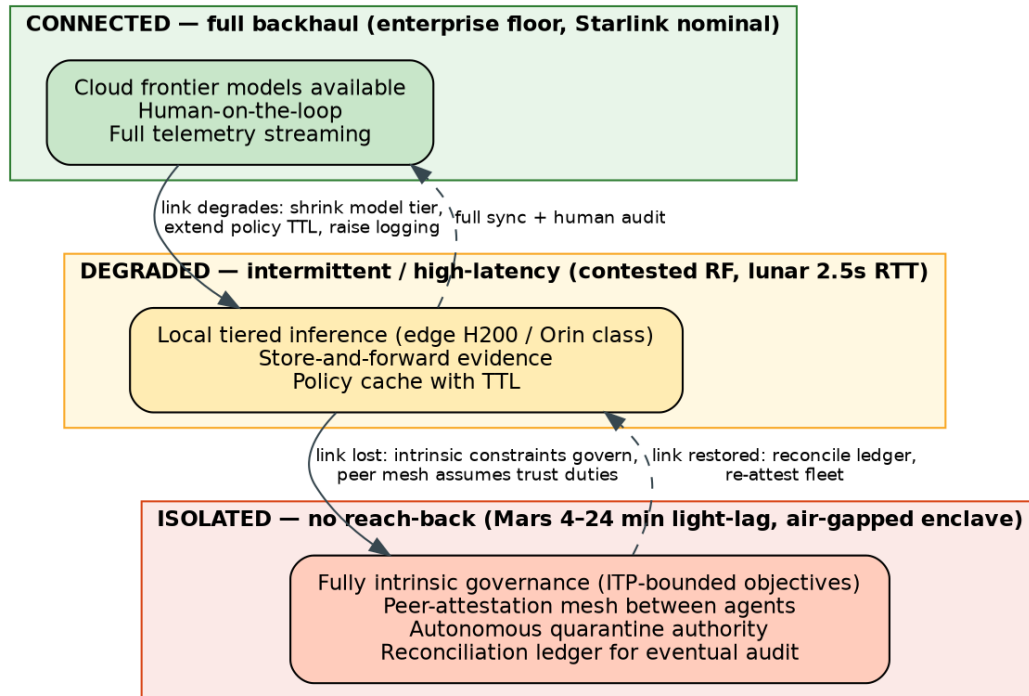


Figure 3 — Connected / Degraded / Isolated tiers and transition behaviors.

Tier	Trust and control behavior	Representative environments
Connected	Central identity authority and PDP authoritative; cloud frontier models available; human-on-the-loop; streaming telemetry; short credential leases.	Enterprise agentic platforms; SOC copilots; factory floor with nominal backhaul.
Degraded	Edge inference tiers assume cognition (H200/Orin-class); policy cache with TTL governs; store-and-forward evidence; leases lengthen; escalations queue.	Contested RF; maritime; subterranean; lunar operations (~2.5 s RTT).
Isolated	Intrinsic constitutional constraints are the sole governance; peer-attestation mesh assumes identity/hygiene duties; autonomous quarantine authority; reconciliation ledger accrues for post-hoc audit.	Air-gapped sovereign enclaves; Martian surface operations (4–24 min light-lag); disaster-degraded infrastructure.

6.1 Transition mechanics

- Downgrade (link loss): model tier shrinks to local; policy TTLs extend under signed degradation rules; logging verbosity rises; kinetic action thresholds tighten (higher confidence required for irreversible acts).

- Upgrade (link restore): fleet-wide re-attestation before privileges restore; reconciliation ledger replays to the flight recorder; human authority reviews quarantine decisions taken autonomously; policy cache refreshes and TTLs reset.

7. Threat Model

The AATA threat model enumerates agent-fleet attack classes and maps each to STRIDE categories and MITRE ATLAS techniques, with the primary mitigating plane. This table generalizes the agentic overlay threat-model pattern, including Unicode steganographic covert channels.

Threat class	STRIDE	ATLAS mapping	Plane	Primary controls
Inter-agent covert channels (Unicode stego, zero-width, timing)	Information Disclosure	AML.T0056 (exfil via ML artifacts); custom TTP: inter-agent stego	Observability	CCM inspection, canonicalization at proxies, traffic normalization
Goal hijacking / indirect prompt injection via poisoned context	Tampering, EoP	AML.T0051 (LLM prompt injection)	Control	Context provenance tags, PDP tool gating, constitutional constraints
Capability escalation through delegation chains	Elevation of Privilege	AML.T0053 (LLM plugin compromise) analog	Trust	Monotone token attenuation, max delegation depth, provenance tracking
Memory / RAG store poisoning (persistent)	Tampering	AML.T0020 (poison training data) runtime analog	Trust + Obs.	Signed corpus manifests, write-path attestation, retrieval provenance
Model supply chain (weights, adapters, skills)	Tampering, Spoofing	AML.T0010 (ML supply chain compromise)	Trust	Stage-2 image attestation, signed manifests, staged rollout rings
Agent impersonation / rogue agent joins fleet	Spoofing	ATT&CK T1078 analog for agents	Trust	Hardware-rooted SVIDs, peer-attestation mesh, mTLS everywhere
Fleet monoculture exploitation (one exploit, all units)	EoP, DoS	AML.T0043 (craft adversarial data) at scale	Governance	P7 baseline diversity, rollout rings, blast-radius caps in tokens
Evidence tampering / decision-log repudiation	Repudiation	ATT&CK T1070 analog	Observability	Hash-chained flight recorder, pre-actuation write, external anchor
Kinetic misuse of embodied actuation	EoP (physical)	ATLAS physical-impact extensions	Control + Exec.	Fail-closed PDP for kinetic classes, reflex-layer safety interlocks independent of cognition
Denial of hygiene (suppress quarantine/telemetry)	DoS, Tampering	ATT&CK T1562 (impair defenses) analog	Control	Hygiene as peer-mesh function, watchdog attestation, dead-man defaults

8. Control Catalog (Baseline)

Minimum viable control set for a v1 deployment. IDs are stable references for client adaptations, POA&Ms, and Jira decomposition.

Control ID	Control	Plane	Standards mapping
AATA-GV-01	Signed constitutional policy store with TTL distribution	Governance	NIST AI RMF GOVERN; 800-53 PL-1
AATA-GV-02	Human authority interface with override and reconciliation audit	Governance	AI RMF MANAGE; 800-53 AC-6(9)
AATA-TR-01	Per-agent hardware-rooted identity (SPIFFE/SVID)	Trust	SP 800-207 (ZTA); IA-2/IA-5
AATA-TR-02	Model + runtime attestation (weights/adapters/prompts digests)	Trust	SSDF; SLSA-style provenance
AATA-TR-03	Scoped capability tokens with monotone delegation attenuation	Trust	800-207; AC-3, AC-4
AATA-CT-01	PDP gating on all tool/actuator calls; fail-closed kinetic	Control	800-53 AC-3(8); IEC 61508 concepts
AATA-CT-02	Autonomous hygiene: graduated quarantine and golden re-image	Control	IR-4; CM-2 golden baseline
AATA-CT-03	Tiered model routing with cost/latency/capability policy	Control	AI RMF MAP/MEASURE
AATA-OB-01	Hash-chained pre-actuation flight recorder	Observability	AU-9, AU-10 (non-repudiation)
AATA-OB-02	Covert-channel monitoring and canonicalization at proxies	Observability	SI-4; ATLAS detection guidance
AATA-OB-03	Behavioral baselining and fleet anomaly analytics	Observability	SI-4(2); AI RMF MEASURE
AATA-EX-01	Mediated tool/actuator proxies; no direct tool access	Execution	AC-4 information flow enforcement
AATA-EX-02	Reflex-layer safety interlocks independent of cognitive layer	Execution	ISO 10218 / functional safety pattern
AATA-DD-01	Defined DDIL behavior for every control (TTL, failover, ledger)	Cross-plane	CP-2 contingency; ZTA offline pattern

9. Deployment Footprints

Footprint	Composition	Reference implementation
F1 — Enterprise Connected	All five planes as platform services; cloud + on-prem hybrid inference; human-on-the-loop SOC integration; short leases.	Agentic overlay on an enterprise SIEM/SOAR estate; Azure hub-spoke with PDP as policy sidecar.
F2 — Sovereign Edge	Trust and Control planes on-prem; local inference cluster (H200-class) as cognition tier; Degraded-mode defaults; store-and-forward evidence.	On-prem GPU inference platform with attested model registry; DoD/regulated-industry enclave pattern.
F3 — Isolated Autonomy	Planes collapse onto the fleet: constitutional constraints on-agent, peer-attestation mesh, reconciliation ledger; Orin-class per-agent compute.	OpenClaw-pattern multi-agent node as testbed; design case for lunar/Martian and air-gapped operations.

10. Implementation Roadmap

Phase	Horizon	Milestones
Phase 0	0–90 days	Threat model instantiation (adopt Section 7 register); flight recorder + tool-proxy mediation on existing agent estate; capability-token schema v1.
Phase 1	3–9 months	Per-agent identity with attestation (TR-01/02); PDP gating with fail-closed kinetic classes; covert-channel monitor deployed at proxies; behavioral baselining online.
Phase 2	9–18 months	Autonomous hygiene (graduated quarantine, golden re-image); Degraded-mode operation certified (policy TTLs, store-and-forward evidence); rollout-ring diversity for fleet baselines.
Phase 3	18–36 months	Isolated-mode certification: intrinsic constitutional runtime, peer-attestation mesh, reconciliation ledger with post-hoc human audit; embodied-fleet extension (reflex interlocks, kinetic PDP classes).

11. Asset Alignment and Positioning

AATA is a generalization layer over existing, proven artifacts. The mapping below identifies what already exists, what it becomes inside the framework, and the go-to-market surface it supports.

Existing asset	AATA role	Positioning surface
Agentic overlay security architecture standard	F1 reference implementation; source of Governance/Control plane specs and the living threat register	Enterprise CISO briefings; critical-infrastructure agentic security engagements
Unicode stego covert-channel threat model	Threat class #1 and the CCM (AATA-OB-02) detection specification	Publishable research note; ATLAS community contribution; differentiated detection IP
H200 on-prem inference platform (BOM + decision matrix)	F2 Sovereign Edge cognition tier; Degraded-mode reference hardware	Federal/regulated sovereign-AI infrastructure consulting
OpenClaw multi-agent framework (tiered routing, token economics)	F3 Isolated Autonomy testbed; Mission Orchestrator (AATA-CT-03) reference design	Living demo: fleet orchestration under budget and comms constraints
ITP / Bounded ITP research	Governance plane theory base — intrinsic constraint architecture for Isolated mode (P1)	DARPA/NSF DDIL-autonomy and AI-assurance funding lines; whitepaper track
Agentic red team methodology	AATA validation harness: adversarial evaluation of each control against Section 7 classes	Productized agentic red-team offering; assurance evidence for clients

11.1 Strategic sequencing

- **Now:** publish AATA v1.0 as the umbrella framework; retrofit existing agentic overlay deliverables as 'AATA F1 instantiation' — one framework, many engagements.
- **Next:** DDIL-autonomy framing for DoD programs (funded today, years before commercial space needs it); AATA-OB-02 covert-channel detection as the technical wedge.
- **Later:** embodied-fleet extension (robot IAM, attestation, kinetic PDP) as the product line that meets the Optimus-class enterprise deployment wave.

Appendix A — Glossary

Term	Definition
AATA	Autonomous Agent Trust Architecture — this framework.
Capability token	Short-lived credential enumerating an agent's permitted tools, data domains, actuation classes, budgets, and delegation depth.
Constitutional policy	Machine-readable bounded objectives and prohibitions compiled into agent runtime; the intrinsic-alignment artifact (ITP pattern).
DDIL	Denied, Degraded, Intermittent, Limited — connectivity conditions the architecture must survive.
Flight recorder	Hash-chained, append-only, pre-actuation decision log; evidentiary system of record.
Monotone attenuation	Rule that a sub-agent's capabilities are always a strict subset of its parent's.
Peer-attestation mesh	Fleet-internal mutual verification assuming Trust-plane duties when the central authority is unreachable.
Reconciliation ledger	Isolated-mode evidence accumulator replayed to the flight recorder and human authority on reconnect.
Reflex agent	Low-latency (System 1) control loop with safety interlocks independent of the cognitive layer.
SVID	SPIFFE Verifiable Identity Document — per-workload cryptographic identity.

Appendix B — Standards and References

- MITRE ATLAS — Adversarial Threat Landscape for AI Systems (technique mappings, Section 7).
- MITRE ATT&CK — enterprise technique analogs for agent-fleet operations.
- STRIDE — threat categorization model (Microsoft).
- NIST AI Risk Management Framework 1.0 — GOVERN/MAP/MEASURE/MANAGE alignment.
- NIST SP 800-207 — Zero Trust Architecture (identity and PDP patterns).
- NIST SP 800-53 rev5 — control mappings in Section 8.
- SPIFFE/SPIRE — workload identity issuance pattern.
- SLSA / NIST SSDF — supply-chain provenance for model artifacts.
- ISO 10218 / IEC 61508 — functional safety patterns for embodied actuation.

Working draft for internal strategy and client adaptation. Instantiations for specific clients inherit their program-level classification and change control.